
Section C

You are advised to spend no more than **65 minutes** on this section.

Enter your answers to **Section C** in your Electronic Answer Document. You **must save** this document at regular intervals.

These questions require you to load the **Skeleton Program** and to make programming changes to it.

0	8
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This question refers to the subroutine `GetHowLongToRun`.

What you need to do:**Task 1**

Modify the subroutine `GetHowLongToRun` so that it checks that the value entered by the user is an integer between -1 to 5 inclusive.

If an invalid value, including a non-integer value, is entered for the duration of the simulation the program should output:

`Invalid input`

The subroutine should not return any values until a valid integer has been entered.

Task 2

Test that the changes you have made work by conducting the following test:

- start the simulation
- enter the value -2
- enter the value 6
- enter the value `w`
- enter the value 1.4
- enter the value 0 (zero).

Evidence that you need to provide

Include the following in your Electronic Answer Document.

0	8	.	1
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Your amended PROGRAM SOURCE CODE for the subroutine `GetHowLongToRun`.

[5 marks]

0	8	.	2
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SCREEN CAPTURE(S) showing the requested test.

[1 mark]

0 9

This question will extend the functionality of the simulation program.

An additional result is required from the simulation program. It is to calculate the percentage of `Field` cells occupied by plants.

The percentage is calculated by dividing the number of cells occupied by plants by the total number of cells and multiplying by 100.

What you need to do:

Task 1

Modify the subroutine `CountPlants` to calculate **and** display the percentage (rounded to the nearest whole number) of the field with plants.

Task 2

Test that your amended program code works by conducting the following test:

- start the simulation
- enter 1 (one year)
- enter Y (load a data file)
- enter the filename `TestCase.txt`

Evidence that you need to provide

Include the following in your Electronic Answer Document.

0 9

1

Your amended PROGRAM SOURCE CODE for the subroutine `CountPlants`.
[2 marks]

0 9

2

SCREEN CAPTURE(S) showing the `Field` contents **and** messages displayed for **spring**. Data for the other seasons should not be included in the evidence.
[1 mark]

Turn over for the next question

Turn over ►

1 0

This question will extend the simulation program to allow `Field` data to be saved to a text file.

When the simulation has run for the required number of years, the user should be given the option of saving the `Field` data to a text file.

This text file must represent the data in the same format as the file `TestCase.txt` stores it (see **Figure 7**), including the numbers at the end of each line.

Figure 7

```

.....S..... | 0
.....X..... | 1
..... | 2
.....SSSSSXSSSSSSSSSS...S..... | 3
..... | 4
.....S.S.....S.S..... | 5
.....S.....S..... | 6
.....S.SSSSSSSSXSS.S..... | 7
.....S.S.....S.S..... | 8
..X..... | 9
.....S.S.S.....S.S.S..... | 10
.....S.X.S.SSSSS.S.S.S..... | 11
.....S.S.S.S...S.S.S.S...X..... | 12
.....S.S.S.S.S.S.S.S.S..... | 13
.....S.S.S.S...S.S.S.S..... | 14
.....S.....SSSSS.....S..... | 15
.....S.S.S.....S.S.S..... | 16
.....XX.....S..... | 17
.....S.S.....S.S..... | 18
.....S.....S..... | 19

```

What you need to do:

Task 1

Create a new subroutine `SaveToFile`. This subroutine is to ask the user whether they want to save the data by displaying the message:

Save the current `Field` state to a text file? (Y/N):

If the user chooses to save the data, the subroutine is to create a new text file with a user-specified filename and store the formatted data in it.

Task 2

In the subroutine `Simulation` add a call to the subroutine `SaveToFile` in an appropriate place.

Task 3

Test that the subroutine `SaveToFile` works by conducting the following test:

- run the simulation for **two** years
- do not load a data file
- save the data to file `Test1.txt`

Task 4

- run the simulation for **one** year
- load the data file `Test1.txt`

Evidence that you need to provide

Include the following in your Electronic Answer Document.

1	0	.	1
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Your PROGRAM SOURCE CODE for the subroutines `SaveToFile` and `Simulation`.

[9 marks]

1	0	.	2
---	---	---	---

For the first run of the simulation: SCREEN CAPTURE(S) showing the `Field` displayed for **winter year 2** including the user interaction for saving the data.

[1 mark]

1	0	.	3
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For the second run of the simulation: SCREEN CAPTURE(S) showing the `Field` displayed for **spring year 1** including the user interaction for loading the data file.

[1 mark]

Turn over for the next question

Turn over ►

1	1
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This question will further extend the functionality of the simulation program.

In the autumn, when plants drop their seeds, there may be a prevailing wind that shifts the seeds so they land in a different position.

If there is **no wind** the seeds land as shown in **Figure 8**.

Figure 8

```

. . . . .
. S S S .
. S P S .
. S S S .
. . . . .

```

If there is a prevailing wind from the **east** the seeds will land as shown in **Figure 9**.

Figure 9

```

. . . . .
S S S . .
S . P . . ←
S S S . .
. . . . .

```

If there is a prevailing wind from the **northwest** the seeds will land as shown in **Figure 10**.

Figure 10

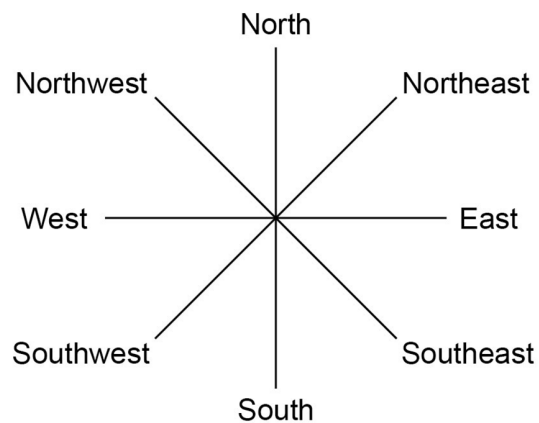
```

      ↘
. . . . .
. . . . .
. . P S S
. . S . S
. . S S S

```

The other wind directions as shown in **Figure 11** have similar effects on where the seeds land.

Each wind direction should occur randomly with each wind direction (or no wind at all) occurring with a 1 in 9 chance.

Figure 11**What you need to do:****Task 1**

Modify the subroutine `SimulateAutumn` to change the position where the seeds land, depending on the prevailing wind direction.

Include code to output a message giving the name of the wind direction that was generated randomly. If there was no wind an appropriate message should be displayed.

Task 2

Test that your amended program code works by conducting the following test.

Without loading a data file run the simulation in stepping mode until **two** different wind directions have occurred.

Evidence that you need to provide

Include the following in your Electronic Answer Document.

1 **1** . **1**

Your amended PROGRAM SOURCE CODE for the subroutine `SimulateAutumn`.

[12 marks]

1 **1** . **2**

SCREEN CAPTURE(S) showing the `Field` contents **and** messages displayed for two autumns with different prevailing wind directions. Data for the other seasons does not need to be included in the evidence.

[1 mark]

END OF QUESTIONS